IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Problem Statement : AI-powered Digital Financial Literacy Agent

The Challenge: Digital financial tools are expanding rapidly, but many users struggle to understand and navigate essential financial practices like using UPI safely, avoiding online scams, managing budgets, and understanding loan interest rates. Reliable financial information is often scattered across complex government portals, banking websites, and educational platforms. Without accessible, personalized, and culturally inclusive guidance in local languages, individuals risk financial fraud and lack the confidence to participate fully in digital finance.

The Objective: Build an **AI-powered Digital Financial Literacy Agent** that retrieves reliable content to help users navigate financial tools, protect against fraud, and build confidence through personalized, multilingual guidance.

Core Requirements / Features:

Multilingual Support: Enable users from diverse linguistic backgrounds to interact and ask questions in their preferred language (e.g., *"How do I send money via UPI?"* or *"What is a safe interest rate for a loan?"*).

Financial RAG Retrieval**:** Fetch verified content regarding UPI usage, scam prevention, budgeting, and personal finance management from trusted government and banking platforms.

Fraud Protection & Advisory: Educate users on identifying phishing links, fake customer care numbers, and safe digital transaction habits.

Multi-Agent System:

Financial Knowledge Agent (retrieves facts on banking and schemes via RAG).

Scam Prevention & Security Agent (analyzes texts/scenarios for potential fraud risks).

Budgeting & Advisory Agent (provides personalized savings and loan guidance).

Interactive Dashboard**:** Visualize financial literacy progress, budgeting tips, and scam alerts.

Proposed Solution -

Proposed Solution: AI-powered Digital Financial Literacy Agent
The proposed solution is an AI-powered Digital Financial Literacy Agent built using Python, Streamlit, IBM watsonx.ai, and Granite Models to provide accessible, personalized, and culturally inclusive financial guidance. The system tackles challenges in digital adoption, scam awareness, and personal finance through a modular, custom-coded architecture.

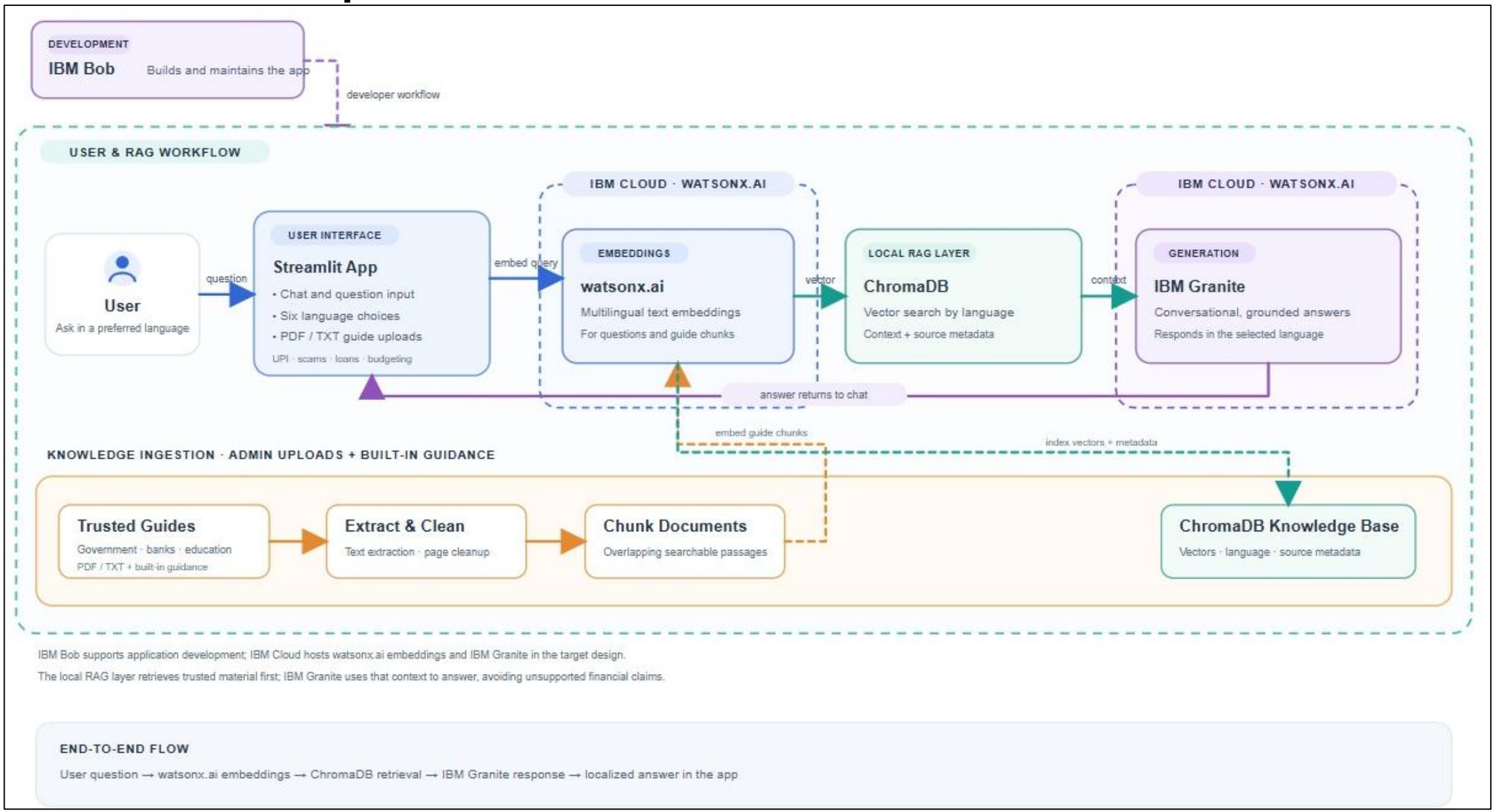
At its core, the solution utilizes a Retrieval-Augmented Generation (RAG) pipeline to fetch reliable guidelines from trusted sources such as RBI, NPCI, and government financial portals. This data is embedded and stored in a vector database, allowing Granite models to retrieve and summarize rules on UPI, interest rates, and budgeting

The system comprises specialized components: a retrieval engine handles policy and tool queries, an analysis module detects online scam patterns, and an advisory interface assists with personal finance management. Streamlit powers the interactive frontend interface, while watsonx.ai ensures enterprise-grade model deployment, embeddings, and governance

Technology Used

Python: Core programming language used for building the application logic, data ingestion, and RAG pipeline.
Streamlit: For building the interactive, clean, and user-friendly web application interface.
IBM Granite Model : (ibm-granite-3-2-8b or Granite-4.0-8B-Instruct)For natural language understanding, complex financial reasoning, scam evaluation, and multilingual personalization.
IBM Cloud: Provides a secure, scalable, and reliable infrastructure environment for deploying the Digital Financial Literacy Agent.
RAG (Retrieval-Augmented Generation): To fetch accurate real-world data from banking and regulatory sources before generating responses, ensuring trustworthiness.
IBM watsonx.ai: For model deployment, vector embeddings, and robust AI governance.
Vector Database (FAISS / Chroma): For fast, efficient RAG-based retrieval of financial regulations, scam alerts, and educational articles.

Architecture blueprint



Role of Agentic AI in the solution

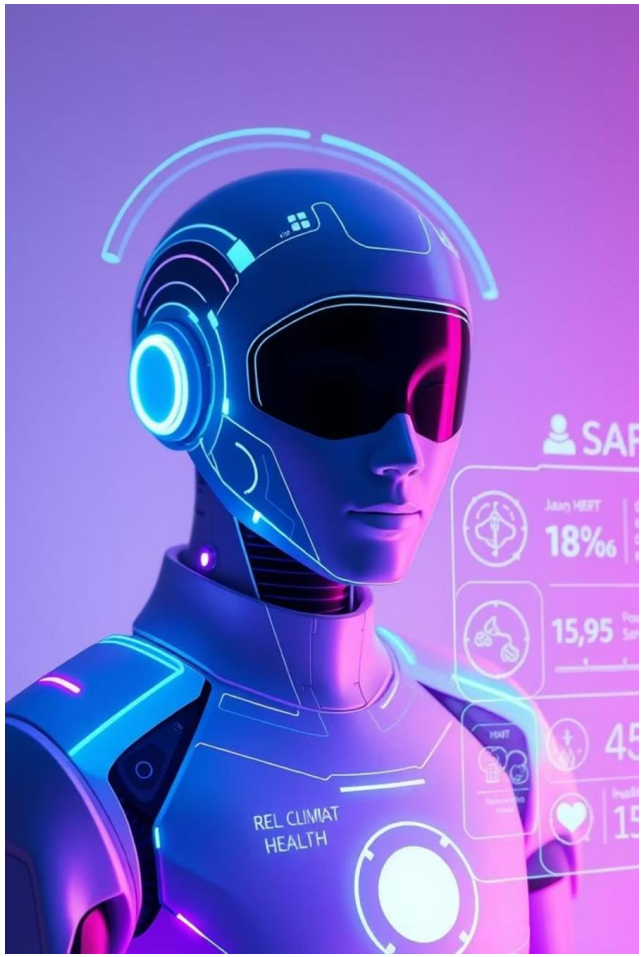
Role of Agentic AI in : AI-powered Digital Financial Literacy Agent

Agentic AI elevates the Digital Financial Literacy Agent from a static chatbot into an autonomous, proactive financial mentor:

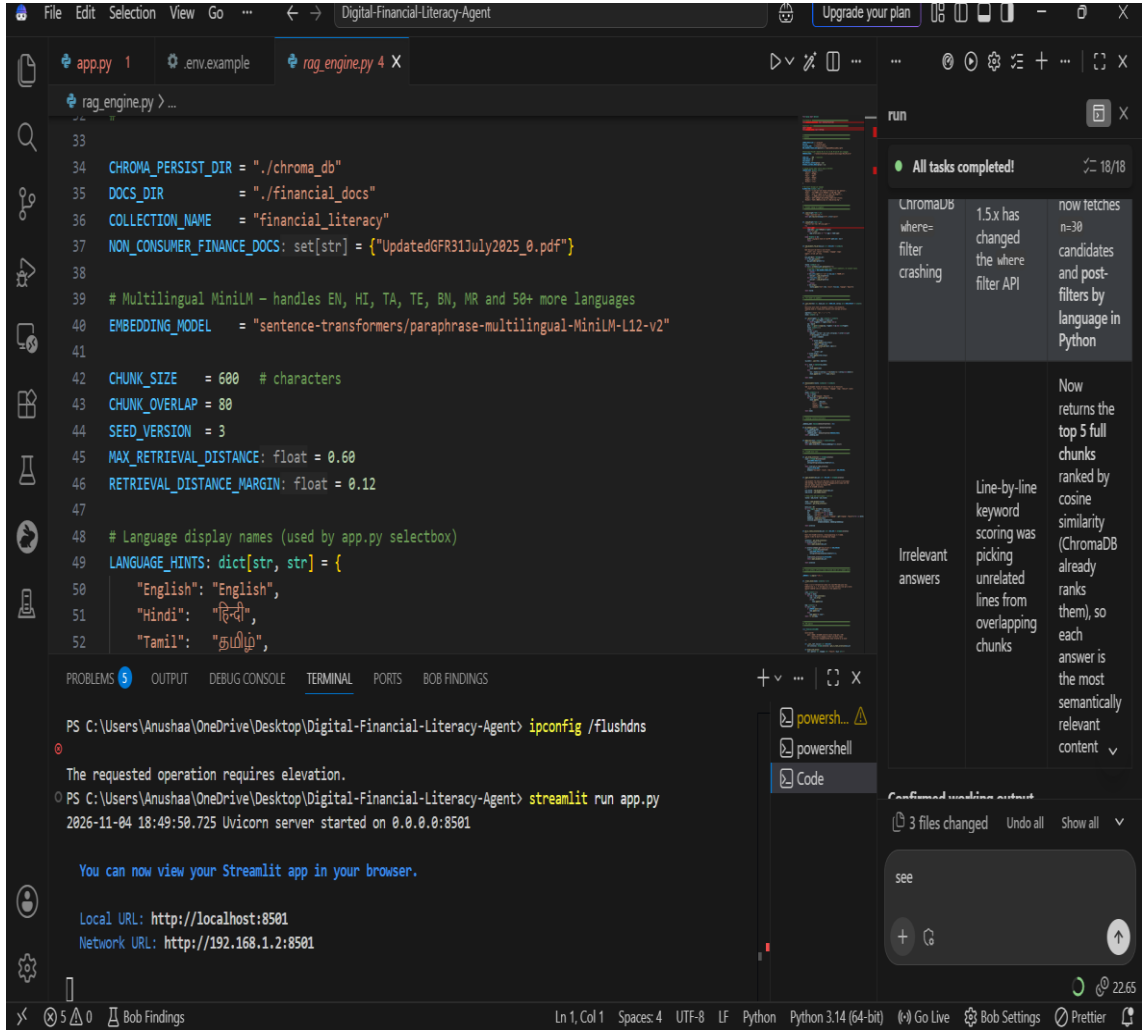
Autonomous Intent Routing: Dynamically evaluates whether a user inquiry requires step-by-step technical execution (like UPI setup), scam risk verification (analyzing suspicious text messages), or budgeting advice, routing it to the appropriate context.

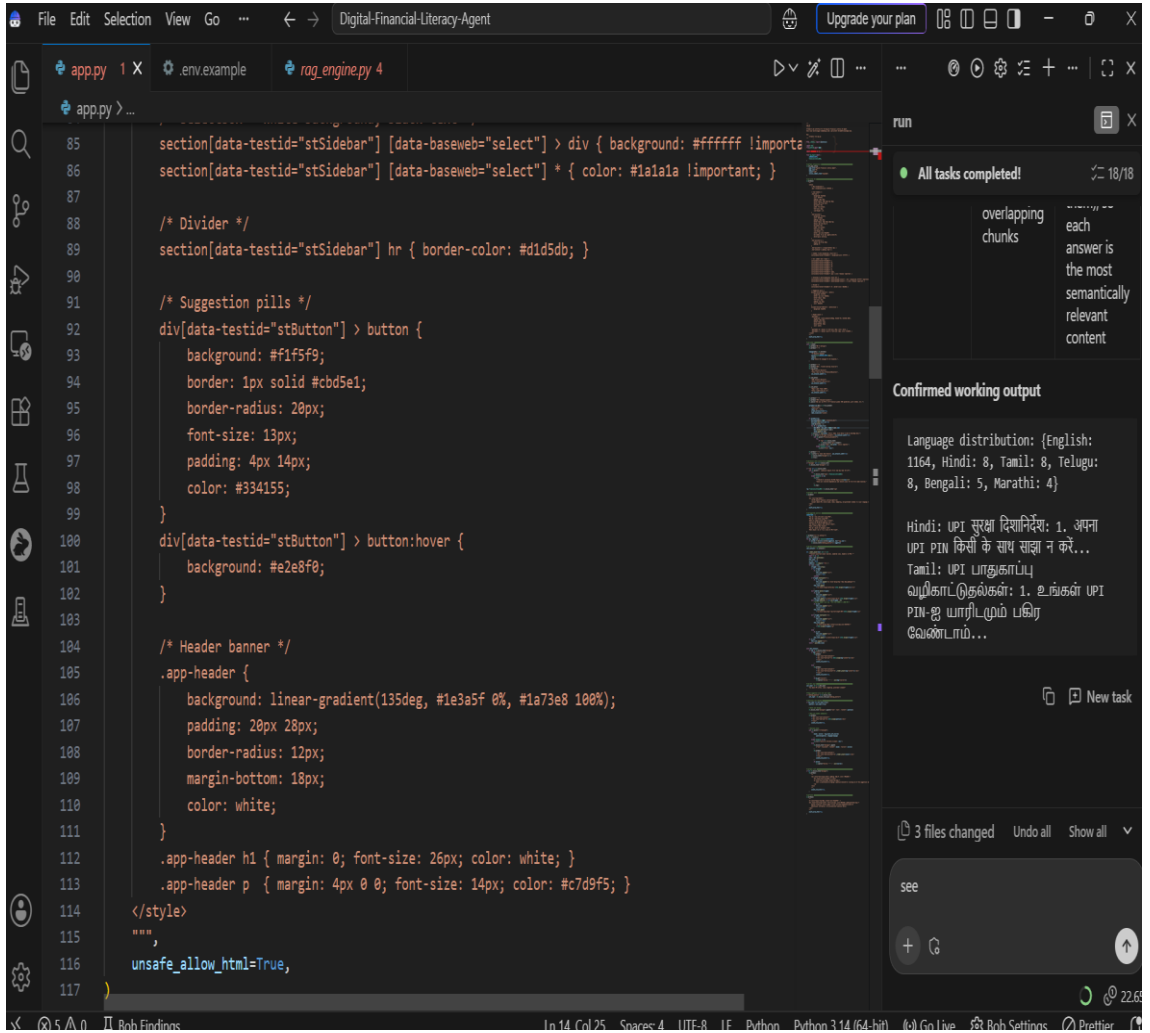
Proactive Guardrails & Verification: Cross-references user queries against verified regulatory data via RAG to ensure that no incorrect or high-risk financial advice is ever delivered.

Context Adaptation & Personalization: Adjusts explanation depth, tone, and regional language nuances based on user interactions, ensuring financial concepts are easily digested by diverse demographic groups.



Snapshots


File Edit Selection View Go
Digital-Financial-Literacy-Agent
Upgrade your plan
app.py 1
.env.example
rag_engine.py 4
rag_engine.py > ...
CHROMA_PERSIST_DIR = "./chroma_db"
DOCS_DIR = "./financial_docs"
COLLECTION_NAME = "financial_literacy"
NON_CONSUMER_FINANCE_DOCS: set[str] = {"UpdatedGFR31July2025_0.pdf"}
Multilingual MiniLM — handles EN, HI, TA, TE, BN, MR and 50+ more languages
EMBEDDING_MODEL = "sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2"
CHUNK_SIZE = 600 # characters
CHUNK_OVERLAP = 80
SEED_VERSION = 3
MAX_RETRIEVAL_DISTANCE: float = 0.60
RETRIEVAL_DISTANCE_MARGIN: float = 0.12
Language display names (used by app.py selectbox)
LANGUAGE_HINTS: dict[str, str] = {
"English": "English",
"Hindi": "हिन्दी",
"Tamil": "தமிழ்",
run
All tasks completed!
18/18
ChromaDB where= filter crashing
1.5.x has changed the where filter API
now fetches n=30 candidates and post-filters by language in Python
Irrelevant answers
Line-by-line keyword scoring was picking unrelated lines from overlapping chunks
Now returns the top 5 full chunks ranked by cosine similarity (ChromaDB already ranks them), so each answer is the most semantically relevant content
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS BOB FINDINGS
PS C:\Users\Anushaa\OneDrive\Desktop\Digital-Financial-Literacy-Agent> ipconfig /flushdns
The requested operation requires elevation.
PS C:\Users\Anushaa\OneDrive\Desktop\Digital-Financial-Literacy-Agent> streamlit run app.py
2026-11-04 18:49:50.725 Uvicorn server started on 0.0.0.0:8501
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://192.168.1.2:8501
powersh...
powershell
Code
3 files changed
Undo all
Show all
see
Bob Findings
Ln 1, Col 1
Spaces: 4
UTF-8
LF
Python
Python 3.14 (64-bit)
Go Live
Bob Settings
Prettier


File Edit Selection View Go
Digital-Financial-Literacy-Agent
Upgrade your plan
app.py 1
.env.example
rag_engine.py 4
app.py > ...
section[data-testid="stSidebar"] [data-baseweb="select"] > div { background: #ffffff !importa
section[data-testid="stSidebar"] [data-baseweb="select"] * { color: #1a1a1a !important; }
/* Divider */
section[data-testid="stSidebar"] hr { border-color: #d1d5db; }
/* Suggestion pills */
div[data-testid="stButton"] > button {
background: #f1f5f9;
border: 1px solid #cbd5e1;
border-radius: 20px;
font-size: 13px;
padding: 4px 14px;
color: #334155;
}
div[data-testid="stButton"] > button:hover {
background: #e2e8f0;
}
/* Header banner */
.app-header {
background: linear-gradient(135deg, #1e3a5f 0%, #1a73e8 100%);
padding: 20px 28px;
border-radius: 12px;
margin-bottom: 18px;
color: white;
}
.app-header h1 { margin: 0; font-size: 26px; color: white; }
.app-header p { margin: 4px 0 0; font-size: 14px; color: #c7d9f5; }
</style>
"""
unsafe_allow_html=True,
run
All tasks completed!
18/18
overlapping chunks
each answer is the most semantically relevant content
Confirmed working output
Language distribution: {English: 1164, Hindi: 8, Tamil: 8, Telugu: 8, Bengali: 5, Marathi: 4}
Hindi: UPI सुरक्षा दिशानिर्देश: 1. अपना UPI PIN किसी के साथ साझा न करें...
Tamil: UPI பாதுகாப்பு வழிகாட்டுதல்கள்: 1. உங்கள் UPI PIN-ஐ யாரிடமும் பகிர வேண்டாம்...
New task
3 files changed
Undo all
Show all
see
Bob Findings
Ln 14, Col 25
Spaces: 4
UTF-8
LF
Python
Python 3.14 (64-bit)
Go Live
Bob Settings
Prettier

Novelty and Uniqueness

- **Zero-Hallucination Grounding via RAG:** Unlike generic financial chatbots that risk making up banking rules, every output is strictly tethered to verified RBI, NPCI, and government educational repositories.
- **Interactive Scam Risk Evaluation:** Features a specialized scanning mechanism where users can paste suspicious links, messages, or offers to receive an instant security risk assessment.
- **Native Language Simplification:** Translates complex legal and banking jargon into simple, culturally inclusive terminology across multiple regional languages.
- **Enterprise-Grade Governance:** Built on IBM watsonx.ai with secure governance parameters ensuring data privacy and compliant model deployment.

Git Hub Link

https://github.com/anushaaj/Financial-literacy-Agent

Future Scope

1. **Integration with Real-Time Banking APIs & Cybercrime Feeds:** Connect directly with official cybercrime reporting portals and live threat feeds to instantly flag newly emerging online banking scams and fake customer care numbers.
2. **Voice-First Regional Language Support:** Expand the Streamlit application to include speech-to-text and text-to-speech capabilities, enabling low-literacy users or rural populations to interact naturally using voice commands in their regional languages.
3. **Offline-First Mobile Web Deployment:** Optimize the application for progressive web app (PWA) architecture or lightweight mobile wrappers so users in low-connectivity areas can access basic financial literacy guides offline.

Certificates Earned:



In recognition of the commitment to achieve professional excellence

Getting Started with Artificial Intelligence
IBM SkillsBuild

Anushaa J

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence

Issued on: Oct 23, 2026
Issued by: IBM SkillsBuild

Verify: https://www.credly.com/badges/58287759-5ebd-4aaf-9152-d5e05ef5d9ee

IBM

Certificates Earned:



IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Anushaa J

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 23 Oct 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.